

Customer Retention Analysis

on Data Analytics Using Power BI

A Project Report

Submitted By

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ABSTRACT



Customer retention is essential for business success and sustainable growth. This project analyzes customer data using Business Intelligence techniques to understand customer behavior and identify factors that influence customer loyalty and churn.

Microsoft Power BI is used to clean, analyze and visualize data effectively. The interactive dashboards provide meaningful insights on customer demographics, service usage, contract details, payment methods, tenure, satisfaction and churn status.

The project helps organizations monitor customer performance, identify high-risk customers, reduce churn, improve customer satisfaction and support data-driven decision-making.

INTRODUCTION



- Customer retention is more cost-effective than acquiring new customers.
- Businesses generate large volumes of data from services, transactions and customer interactions.
- This project analyzes customer retention and churn using Power BI dashboards.
- It helps organizations identify high-risk customers and improve retention strategies.
- The analysis includes demographics, tenure, contract type, payment method, service usage and satisfaction.

TECHNOLOGY STACK



Microsoft Power BI
(Visualisation & Reporting)



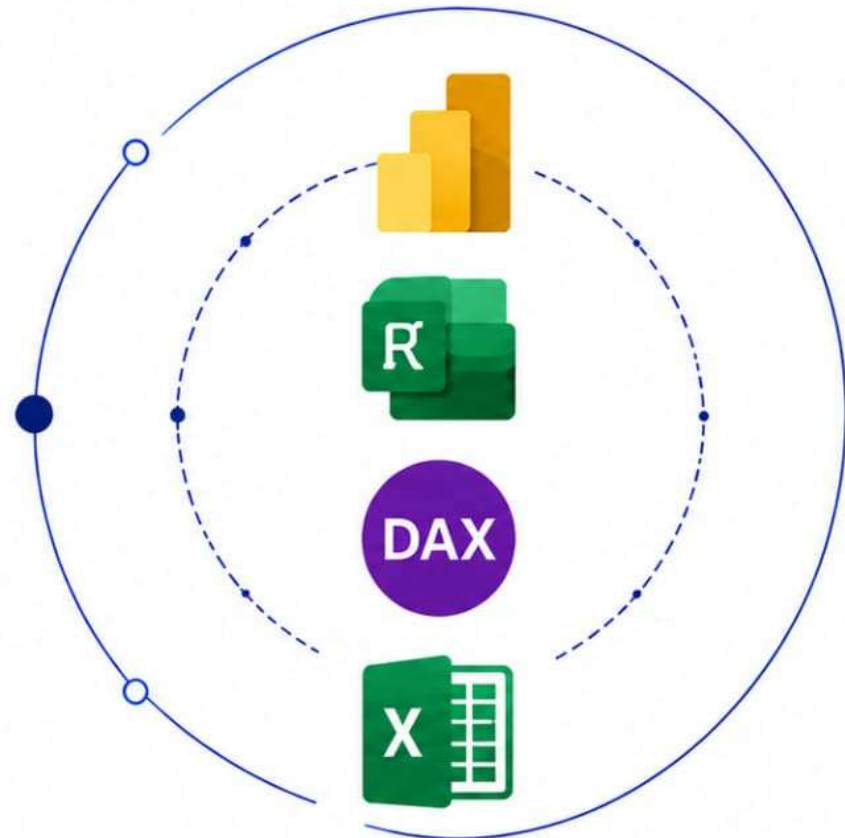
Power Query
(Data Cleaning & Transformation)



DAX
(Calculations & KPIs)

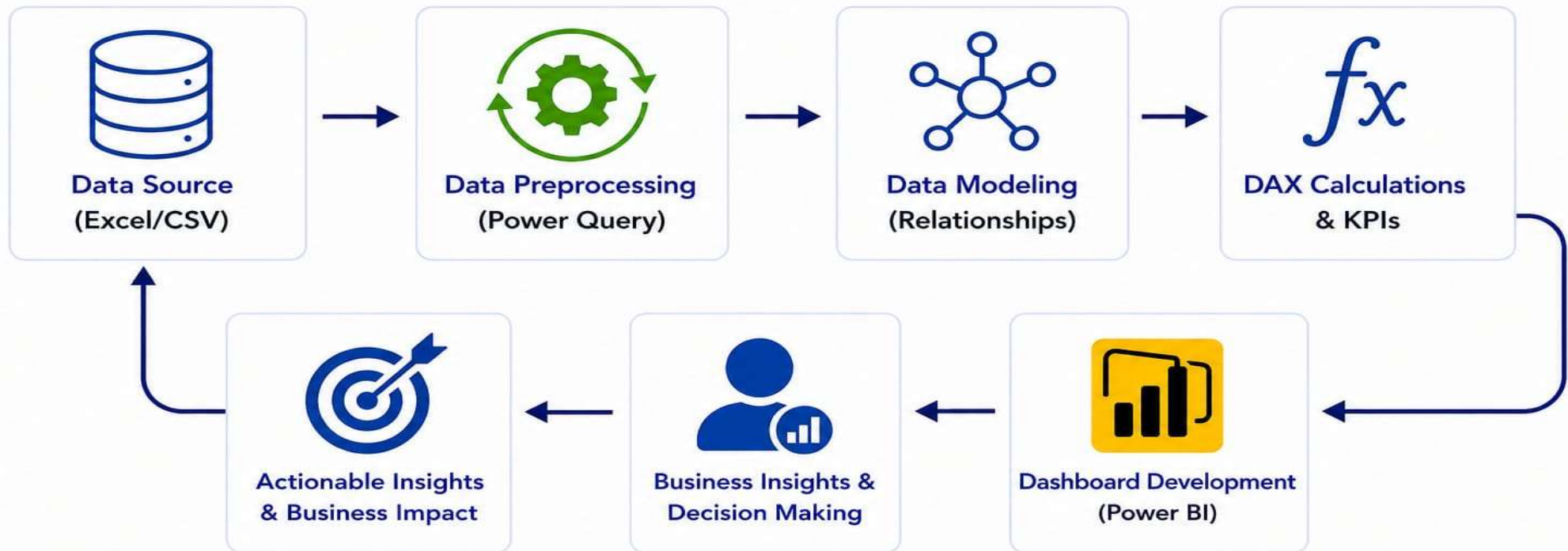


Excel / CSV Dataset
(Telco Customer Churn Data)



SYSTEM DESIGN / ARCHITECTURE

The architecture below shows the end-to-end flow of data from source to insights and decision-making.



This architecture ensures clean data flow, accurate analysis, and interactive dashboards that provide meaningful insights to support data-driven decision-making.

IMPLEMENTATION PROCESS

- 1 Data Collection
- 2 Data Cleaning & Transformation (Power Query)
- 3 Data Modeling
- 4 DAX Measures & KPI Creation
- 5 Dashboard Development (Power BI)
- 6 Customer Retention & Churn Analysis
- 7 Business Insights & Decision Making



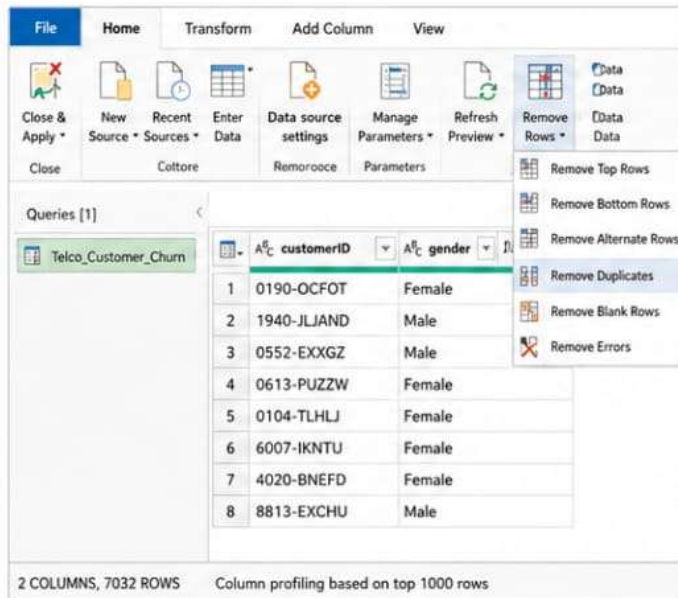
DATASET OVERVIEW

- Source: Telco Customer Churn Dataset
- Records: ~7,032 Rows
- Columns: 21
- Key Features:
 - ✓ Customer Demographics
 - ✓ Tenure, Contract, Payment Method
 - ✓ Service Usage
 - ✓ Monthly Charges & Total Charges
 - ✓ Satisfaction & Churn Status

customerID	gender	SeniorCitizen	partner	Dependents	tenure	Churn	...
0190-OCFOT	Female	0	Yes	1	9	Yes	...
1940-JLJAND	Male	0	No	1	9	No	...
0552-EXXGZ	Male	0	No	1	4	No	...
0613-PUZZW	Female	0	No	1	13	Yes	...
0104-TLHLJ	Female	0	No	1	3	Yes	...
6007-IKNTU	Female	0	No	1	7	No	...
4020-BNEFD	Female	0	No	1	1	No	...
8813-EXCHU	Male	0	No	1	7	Yes	...
6680-THJCF	Male	0	No	1	5	No	...
2702-LVYGI	Female	0	No	1	1	No	...
3826-OBPTH	Male	0	No	1	7	Yes	...
5205-COZLW	Female	0	No	1	1	No	...
9167-CGHLZ	Female	0	No	1	7	Yes	...
5305-GDSLW	Female	0	Yes	0	6	No	...
0690-UOFBG	Female	0	No	1	3	No	...
5545-KAIFB	Male	1	No	0	1	No	...
4795-RUJEK	Female	1	No	0	4	No	...
0741-OKOMC	Male	1	No	0	2	No	...
4397-QRDMV	Male	1	No	0	2	Yes	...
6711-OKOMC	Female	1	Yes	0	2	No	...
7932-OBODW	Female	1	Yes	0	5	No	...

DATA CLEANING – POWER QUERY

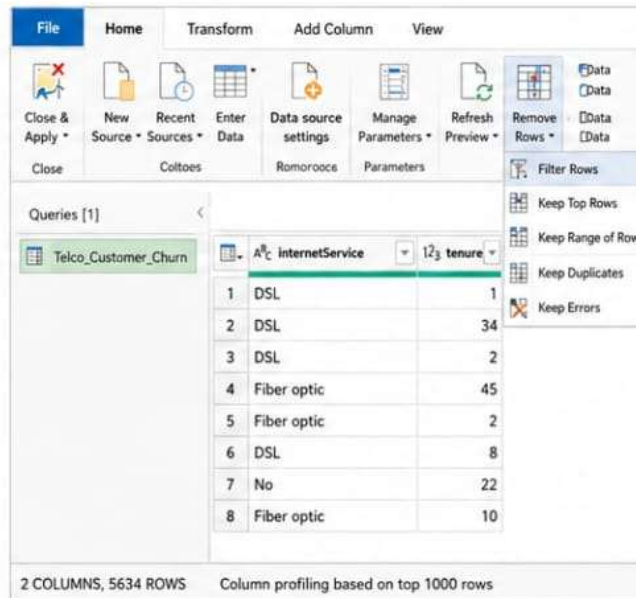
Removed Duplicates



2 COLUMNS, 7032 ROWS Column profiling based on top 1000 rows

	customerID	gender
1	0190-OCFOT	Female
2	1940-JLJAND	Male
3	0552-EXXGZ	Male
4	0613-PUZZW	Female
5	0104-TLHLJ	Female
6	6007-IKNTU	Female
7	4020-BNEFD	Female
8	8813-EXCHU	Male

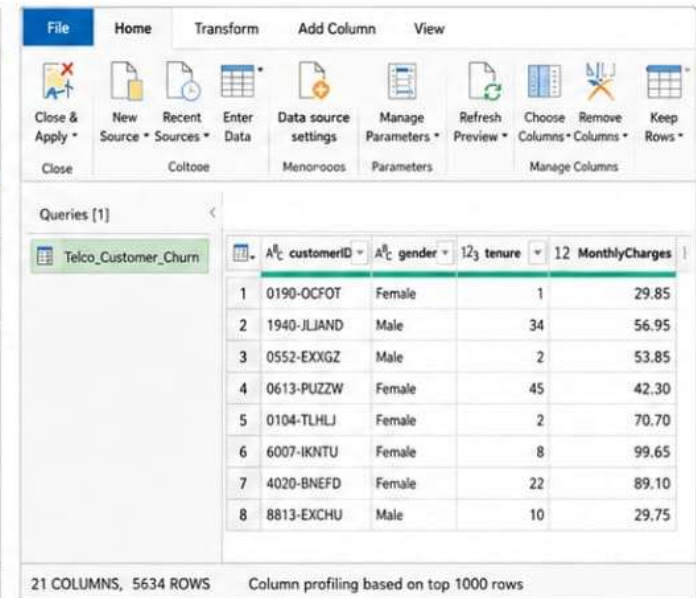
Filtered Rows



2 COLUMNS, 5634 ROWS Column profiling based on top 1000 rows

	InternetService	tenure
1	DSL	1
2	DSL	34
3	DSL	2
4	Fiber optic	45
5	Fiber optic	2
6	DSL	8
7	No	22
8	Fiber optic	10

Cleaned Data



21 COLUMNS, 5634 ROWS Column profiling based on top 1000 rows

	customerID	gender	tenure	MonthlyCharges
1	0190-OCFOT	Female	1	29.85
2	1940-JLJAND	Male	34	56.95
3	0552-EXXGZ	Male	2	53.85
4	0613-PUZZW	Female	45	42.30
5	0104-TLHLJ	Female	2	70.70
6	6007-IKNTU	Female	8	99.65
7	4020-BNEFD	Female	22	89.10
8	8813-EXCHU	Male	10	29.75



Power Query is used to remove duplicates, handle missing values, change data types and transform data for accurate analysis.

KEY TAKEAWAYS



Power BI dashboards help in understanding customer behavior effectively.

Interactive dashboards provide clear visuals and actionable insights for data-driven decision-making.



Churn analysis identifies customers at risk of leaving.

Early identification enables targeted retention strategies to reduce churn and improve loyalty.



Data-driven strategies improve customer satisfaction and loyalty.

Insights from demographics, tenure, services, and payments help personalize offers and enhance experience.



Interactive dashboards provide real-time business insights.

Dynamic visuals allow stakeholders to explore data and make informed decisions quickly.



Better decision-making leads to improved retention and revenue.

Focused retention programs and optimized services drive sustainable business growth.



These key takeaways highlight the value of using Power BI to transform customer data into meaningful insights, enabling organizations to enhance retention, increase revenue, and achieve long-term success.

FUTURE SCOPE

This project lays a strong foundation for advanced analytics and data-driven initiatives that can further enhance customer retention and business growth.



1. Integrate real-time data using cloud databases (Azure / SQL).

Enable real-time data ingestion and updates to monitor customer behavior and retention metrics dynamically.



2. Implement machine learning models to predict churn.

Use classification models to identify customers at risk and take proactive retention actions.



3. Include customer feedback & sentiment analysis.

Analyze feedback and sentiment from surveys, chats, and reviews to understand customer satisfaction drivers.



4. Add more KPIs like CLV, NPS, CCR, Repeat Purchase Rate.

Expand KPI coverage to measure customer lifetime value, advocacy, loyalty and engagement more comprehensively.



5. Develop mobile dashboards in Power BI Service.

Create mobile-optimized dashboards for on-the-go access and data-driven decision-making.



6. Implement row-level security for data access.

Enhance data security and privacy by restricting access to data based on user roles and responsibilities.



These future enhancements will enable the organization to build a more intelligent, scalable, and proactive customer retention strategy, driving higher loyalty, satisfaction, and revenue growth.



CONCLUSION

The Customer Retention Analysis Using Power BI project demonstrates how Business Intelligence and data analytics convert raw customer data into meaningful insights.



Interactive dashboards

The interactive dashboards help organizations understand customer behavior, identify churn patterns, and improve retention strategies.



Data-driven insights

These insights support data-driven decision-making, reduce churn, improve customer satisfaction, and contribute to long-term business growth.



Business impact

By leveraging Power BI, organizations can focus on high-value customers, optimize services, and maximize revenue and customer lifetime value.



Strategic advantage

This project highlights the power of analytics in gaining a competitive edge through better understanding of customers and their needs.



Future readiness

The analysis lays a strong foundation for predictive analytics, automation, and AI-driven customer engagement.



Overall, this project showcases how Power BI can transform customer data into actionable insights, helping organizations enhance retention, increase revenue, and achieve sustainable growth.

Thank You!

Any Questions?



Thank you for your time and attention.